WORK-RELATED INJURIES

IN THE ITALIAN TRANSPORTATION AND STORAGE SECTOR

Data Curation
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Table of Contents

DATA CURATION DOCUMENT 5
Document Control 6
1. Executive Summary 7
2. Data Sources 8
2.1 Overview 8
2.2 Source Authority and Data Reliability 8
2.3 Dataset Inventory 8
2.4 Occupational Injury Data (ESAW Domain) 8
2.5 Employment Data (LFS Domain) 9
2.6 Geographic Coverage 9
2.7 Temporal Coverage 9
2.8 Data Extraction and Format 10
2.9 Relationship Between Sources 10
2.10 License 10
2.11 Summary 10
3. Methodological Approach 11
3.1 Analytical Objective 11
3.2 Sectoral Focus 11
3.2.1 Dual Comparative Framework 11
3.3 Geographic Scope Selection 12
3.4 Temporal Scope 12
3.5 Use of Employment as Exposure Denominator 12
3.6 Integration of Heterogeneous Statistical Sources 13
3.7 Preservation of Statistical Integrity 13
3.8 Analytical Use Cases Supported by the Framework 13
3.9 Methodological Positioning 14
4. Data Preparation 15
4.1 Overview 15
4.2 Data Profiling 15
4.3 Data Cleaning 15
4.4 Structural Harmonization 16
4.5 Treatment of Structural Missing Data 16
4.6 Redistribution of “Unknown” Gender Records 16
4.7 Derived Analytical Indicators 17
4.8 Output of the Preparation Phase 17

4.9 Data Integrity Principles Applied 17
5. Treatment of Data Limitations 18
5.1 Overview 18
5.2 Structural Breaks in Time-Series 18
5.3 Backcasting of Missing Historical Values 18
5.4 Interpretation of Zero Values in Residual Categories 19
5.5 Gender Information Gaps 19
5.6 Incompatibility of Age-Class Structures 19
5.7 Differences in Dataset Dimensionality 19
5.8 Analytical Scope Resulting from These Constraints 20
5.9 Transparency and Traceability Measures 20
6. Database Design 21
6.1 Design Objectives 21
6.2 Schema Architecture 21
6.3 Analytical Grain 21
6.4 Fact Tables 22
6.5 Dimension Tables 22
6.6 Data Standardization and Naming Conventions 22
6.7 Referential Integrity 23
6.8 Derived Measures Stored in the Database 23
6.9 Data Reduction and Column Pruning 23
6.10 Integration Logic 23
6.11 Suitability of the Design for Statistical Analysis 24
6.12 Summary 24
7. Data Model 25
7.1 Conceptual Model 25
7.2 Entity Relationships 25
7.3 Analytical Grain 25
7.4 Integration Logic 25
7.5 Advantages of the Adopted Model 26
8. Data Dictionary 27
8.1 Purpose 27
8.2 Identifier Fields 27
8.3 Measure Fields 27
8.4 Derived Variables 28
8.5 Metadata and Descriptive Fields 28

8.6 Naming Convention 28
9. Reproducibility 29
9.1 Reproducibility Objective 29
9.2 Processing Environment 29
9.3 Transformation Traceability 29
9.4 Version Control and Transparency 29
9.5 Reuse Potential 30
9.6 Use of AI-Assisted Tools 30
10. Limitations 31
10.1 Statistical Constraints 31
10.2 Comparability Restrictions 31
10.3 Interpretation of Estimated Values 31
10.4 Analytical Scope 31
10.5 Dependence on Source Updates 31
Appendix A – EUROSTAT Dataset Inventory 32
A.1 ESAW Detailed Accident Datasets (Circumstance-Level Sources) 32
A.2 ESAW Aggregated Datasets (Totals and Validation) 38
A.3 Employment Datasets Used as Exposure Denominators (LFS Domain) 40
A.4 Italy ESAW Datasets for Cross-Activity Comparison (NACE A–U) 42
Appendix B – Phase 1 Outputs (Curated CSV Layer) 44
Appendix C – Derived Indicator Formulae 54
Appendix D – Phase 2 Outputs (SQLite Database Layer) 55
D1. Overview 55
D2. Dimension tables 56
D3. Fact tables 58
D4. Data Lineage 64
Appendix E – Entity–Relationship Diagram (ERD) 65
Appendix F – Classification Systems Used 66
Appendix G – Excel Transformations (Pivot & Formula Schema) 67
G.1 Pivot Tables (Aggregation Layer) 67
G.2 Lookup & Integration Formulas (Denominator Alignment) 69
G.3 Derived Indicators 71
G.4 Backcasting (Structural Missing Data) 72
G.5 Integrity Controls 73
Appendix H – Data Processing Workflow 74

DATA CURATION DOCUMENT

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Document Control

Version	Date	Author	Description
0.1	Dec. 28, 2025	Giulio Galbiati	Initial document structure created based on the C1 template; defined main sections and analytical scope.
0.2	Jan. 3, 2026	Giulio Galbiati	Compiled and documented data source inventory, including origin, format, and access method.
0.3	Jan. 11, 2026	Giulio Galbiati	Added preliminary methodology description; documented data preparation steps and transformation workflow for reproducibility.
0.4	Feb. 2, 2026	Giulio Galbiati	Defined and documented database schema, including tables, keys, and relationships supporting the analytical model.
0.5	Feb. 20, 2026	Giulio Galbiati	Introduced data dictionary with variable definitions and units; added reproducibility notes and standardized technical terminology.
0.6	Feb. 23, 2026	Giulio Galbiati	Technical review: updated formatting, refined figures, verified cross-references, and aligned document with submission guidelines.
0.7	Feb. 25, 2026	Giulio Galbiati	Final review: verified CSV files, validated Excel Pivot Tables and formula summaries for Phase 1, and reviewed DB table summaries for Phase 2.
1.0	Feb. 26, 2026	Giulio Galbiati	First official release submitted for assessment; content frozen for evaluation.

Document Status:

This document represents the final version of the data curation report prepared for the Amazon Data Analytics Program.

It describes the data sources used, the transformation steps carried out, and the structure developed to organize the data for analysis.

If any updates are needed in the future, they will be issued as a new version and recorded in the Document Control table to keep track of changes.

1. Executive Summary

This document describes the curation, harmonization, and structuring of a statistical dataset on work-related injuries in the Italian Transportation and Storage sector (NACE Rev. 2 – Activity H) for the period 2014–2023.

The project integrates multiple EUROSTAT data sources, including accident records (ESAW domain) and labor-force statistics (LFS domain), into a unified analytical database designed to support cross-country comparison and sector-specific safety analysis. Because raw injury counts are not directly comparable across countries with different workforce sizes, employment data were incorporated to construct normalized indicators expressed as casualties per 1,000 employees.

The curation process addressed several challenges typical of multinational administrative statistics, including structural breaks in time-series, inconsistent dimensional coverage, and partial demographic information. Rather than smoothing or interpolating such inconsistencies, the methodology preserves observed data and explicitly flags any estimated values introduced to maintain aggregate continuity.

The final output is a relational SQLite database organized under a Multi-Star Schema, enabling multidimensional analysis across countries, years, demographic groups, and accident circumstances while maintaining traceability to the original statistical sources.

This curated dataset is intended to support descriptive statistical analysis, comparative risk assessment, and evidence-based discussion of occupational safety patterns in the Transportation and Storage sector. It is not designed for causal inference but for transparent and reproducible analytical reporting grounded in official European statistics.

The primary deliverable is a curated analytical database and documented transformation workflow enabling reproducible descriptive analysis.

2. Data Sources

2.1 Overview

The analytical dataset was constructed using publicly available statistical data extracted from EUROSTAT.
Two principal statistical domains were integrated:

- **European Statistics on Accidents at Work (ESAW)** – providing detailed records of occupational injuries.
- **European Labour Force Survey (LFS)** – providing employment statistics used as the exposure denominator for rate calculations.

The integration of these domains enables the transformation of raw accident counts into normalized indicators that support cross-country comparison.

2.2 Source Authority and Data Reliability

All source files originate from EUROSTAT, the statistical office of the European Union, which ensures:

- Harmonized data collection methodologies across Member States
- Standardized classifications (NACE Rev. 2, demographic variables, severity codes)
- Quality assurance procedures applied at national and European levels
- Public accessibility and reproducibility of extracted datasets

The project does not modify original statistical definitions but restructures them for analytical usability.

2.3 Dataset Inventory

Twelve CSV datasets were extracted and curated. These datasets fall into three functional categories:

Category	Dataset Type	Analytical Role
Detailed accident datasets	Injuries by circumstance, activity, severity, sex, and age	Core analytical numerator
Aggregated accident datasets	Fatal and non-fatal totals	Validation and sector-level aggregation
Employment datasets	Employment by activity, country, and occupation	Exposure denominator

2.4 Occupational Injury Data (ESAW Domain)

The ESAW-derived datasets provide multidimensional information on workplace accidents, including:

- Type of working environment, working process and workstation

- Specific physical activity at the time of the accident
- Deviation or triggering event
- Contact mode of injury
- Severity classification (fatal vs. non-fatal)
- Demographic attributes (sex and age class)
- Economic activity classified under NACE Rev. 2

These datasets enable detailed characterization of how and under what conditions occupational injuries occur within the Transportation and Storage sector.

2.5 Employment Data (LFS Domain)

Employment datasets derived from the Labour Force Survey provide the population-at-risk required to contextualize injury counts.

Variables include:

• Number of employed persons by economic activity (NACE Rev. 2)
• Occupational classifications (ISCO-08)
• Country-level employment totals
• Annual observations aligned with the ESAW reference period

These data serve as the denominator for calculating incidence rates and allow comparison between countries with different workforce sizes.

2.6 Geographic Coverage

Two complementary geographic scopes were used:

- **36 European countries** included in employment datasets to establish a comprehensive labor-force reference.
- **10 countries** selected for detailed injury analysis based on their employment weight in Activity H.

This approach balances analytical depth with statistical robustness, ensuring that the countries analyzed represent the majority of sectoral employment exposure.

2.7 Temporal Coverage

All datasets were restricted to the common observation window **2014–2023**, chosen to maximize overlap between ESAW injury reporting and LFS employment data.

This period provides:

- A continuous ten-year analytical horizon
- Sufficient observations for trend analysis
- Consistency across integrated statistical domains

2.8 Data Extraction and Format

Datasets were downloaded in CSV format using EUROSTAT custom extraction tools with harmonized filters applied to:

- Economic activity (NACE Rev. 2 – Activity H for sectoral analysis)
- Time period (2014–2023)
- Geographic selection
- Exclusion of aggregate "Total" categories when disaggregated analysis was required

The CSV format enabled transparent preprocessing and facilitated reproducible transformation into the relational database environment.

2.9 Relationship Between Sources

The curated database relies on a complementary relationship between the two statistical domains:

- Accident datasets measure **events** (numerator).
- Employment datasets measure **exposure** (denominator).

Their integration allows the computation of standardized safety indicators while maintaining the independence of each original statistical system.

2.10 License

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2.11 Summary

The project does not generate new primary data; rather, it restructures authoritative European statistics into an analytically coherent environment.

By combining ESAW injury records with LFS employment measures, the curated dataset enables statistically grounded comparison of occupational risk within the Transportation and Storage sector while preserving traceability to the original EUROSTAT sources.

3. Methodological Approach

3.1 Analytical Objective

The methodological objective of this project is to construct a statistically coherent dataset that allows comparison of occupational injury incidence across European countries within the Transportation and Storage sector.

Raw accident counts alone do not provide meaningful comparisons because they are strongly influenced by workforce size, reporting practices, and sectoral composition. To address this, the analysis adopts an exposure-based framework in which injury occurrences are evaluated relative to employment levels.

This approach aligns with standard occupational-safety methodologies used in official statistics and risk-rate calculations.

3.2 Sectoral Focus

The analysis is restricted to NACE Rev. 2 Activity H (Transportation and Storage), a sector characterized by:

• High operational risk exposure
• Cross-national logistical integration
• Comparability across EU statistical systems
• Availability of harmonized ESAW reporting variables

Focusing on a single economic sector reduces structural heterogeneity and improves interpretability of cross-country comparisons.

3.2.1 Dual Comparative Framework

The analytical design adopts a dual comparative perspective combining cross-country sector analysis with a within-country cross-activity comparison to contextualize occupational risk in the Italian Transportation and Storage sector:

- Cross-country comparison is conducted exclusively within NACE Rev.2 Activity H, ensuring structural homogeneity when evaluating differences between national safety patterns.
- A within-country cross-activity comparison is conducted exclusively within Italy, allowing the Transportation and Storage sector to be interpreted relative to the broader national economic structure (Activities A–U).

This separation avoids conflating:

- sectoral heterogeneity (which would bias cross-country comparisons), and
- national institutional differences (which would bias cross-sector comparisons).

The approach therefore preserves comparability by holding one analytical dimension constant while varying the other.

This analytical logic is reflected structurally in the database through the separation between HC10 cross-country fact tables and the Italy-specific WRI_IT_TOTALS_AU table.

3.3 Geographic Scope Selection

A subset of ten EU countries was selected based on their relative employment size in Activity H during the observation period.

This selection ensures that:

- Countries included represent the majority of the sector's workforce exposure
- Statistical series are sufficiently robust for longitudinal comparison
- Analytical results are not dominated by small-sample volatility

Employment data covering a broader set of countries were retained to provide contextual reference and denominator validation.

3.4 Temporal Scope

The study period (2014–2023) reflects the longest recent interval with sufficiently consistent ESAW and LFS data overlap.

This ten-year window enables:

- Observation of medium-term safety trends
- Mitigation of short-term reporting anomalies
- Construction of stable comparative indicators

Years with structural discontinuities were preserved but treated explicitly during the data-limitation phase.

3.5 Use of Employment as Exposure Denominator

To allow meaningful comparison of injury risk, employment figures were integrated as the reference population.

The principal analytical indicator derived is:

Casualties per 1,000 Employees

This normalization transforms administrative counts into incidence measures that better approximate relative occupational risk.

Such rate-based indicators are widely used in occupational epidemiology and official statistical reporting.

The normalization calculations were implemented during the data-preparation stage using spreadsheet-based transformation. The relational database subsequently stores these derived indicators and allows their recomputation for validation purposes rather than serving as the primary calculation environment.

3.6 Integration of Heterogeneous Statistical Sources

The project combines datasets with different statistical purposes:

Source Domain	Function in Analysis
ESAW (Accidents at Work)	Event measurement (numerator)
LFS Employment Statistics	Exposure measurement (denominator)

Because these systems were not originally designed for joint analysis, harmonization was required to align classifications, temporal coverage, and dimensional compatibility.

3.7 Preservation of Statistical Integrity

A guiding methodological principle throughout the project was to avoid introducing artificial continuity into official statistics.

Accordingly:

- Observed data were never overwritten.
- Structural gaps were documented rather than interpolated.
- Estimation procedures were applied only when necessary for aggregate coherence and were explicitly flagged.
- Analytical conclusions rely primarily on reported, not reconstructed, values.

This conservative treatment ensures that the curated dataset remains faithful to the statistical meaning of the original sources.

3.8 Analytical Use Cases Supported by the Framework

The resulting dataset is designed to enable:

- Cross-country comparison of injury incidence within a shared sector.
- Within-country cross-activity comparison (Italy)
- Examination of accident circumstances and severity patterns.
- Gender-disaggregated safety analysis.
- Longitudinal descriptive trend assessment.

The framework intentionally avoids causal modeling or predictive inference, emphasizing transparency and comparability over statistical extrapolation.

3.9 Methodological Positioning

Conceptually, the project sits at the intersection of:

- Statistical data integration.
- Occupational risk measurement.
- Dimensional data structuring for statistical analysis.

It reorganizes administrative statistical outputs into a structured environment that supports reproducible analysis while maintaining adherence to official data production standards.

4. Data Preparation

4.1 Overview

The data preparation phase transformed heterogeneous EUROSTAT source files into a harmonized analytical layer suitable for relational modeling and rate-based analysis.

This stage focused on profiling, cleaning, structural harmonization, and spreadsheet-based derivation of normalized indicators prior to database loading, while preserving the integrity of observed statistical data.

All transformations were designed to be transparent, reproducible, and reversible to the original source datasets.

4.2 Data Profiling

Initial profiling was conducted to understand dataset structure, dimensional coverage, and comparability across sources.

Key characteristics identified:

- Observation period: 2014–2023 (10 years)
- Geographic scope:
  - Employment data: 36 EU countries
  - Injury datasets: Top 10 countries by employment in Activity H
- Sectoral focus: NACE Rev. 2 – Transportation and Storage (Activity H)
- Severity classification: Fatal vs. non-fatal accidents
- Demographic variables: Sex and Age Class (with partial inconsistencies across datasets)

Employment data were used as the reference population to normalize injury counts and derive incidence indicators.

4.3 Data Cleaning

Several fields contained no analytical value and were removed to reduce noise and redundancy.

Issue Identified	Treatment Applied	Rationale
Columns with 100% null values (e.g., Time, Observation Value metadata, Confidentiality flags)	Dropped	Non-informative administrative metadata
Duplicated descriptive labels	Standardized	Avoid redundancy in relational model
Inconsistent naming conventions	Harmonized	Ensure schema compatibility

Issue Identified	Treatment Applied	Rationale
Structural zero values	Reclassified as missing (not true zeros)	Represent reporting gaps rather than absence of accidents

No interpolation was performed across structural discontinuities in time-series.

4.4 Structural Harmonization

Datasets originated from different ESAW extraction views and required alignment before integration.

Harmonization actions included:

- Standardization of country selection based on employment weight.
- Alignment of NACE Rev. 2 classifications across injury and employment datasets.
- Consolidation of six accident "circumstance" classifications into a unified categorical structure.
- Verification of attribute uniqueness and dimensional consistency.
- Preservation of original categorical granularity for analytical drill-down.

This step ensured that all datasets could be integrated using shared analytical dimensions: Country, Year, Activity, Severity, Sex, and Circumstance.

4.5 Treatment of Structural Missing Data

Several countries exhibited breaks in ESAW reporting, producing zero values that did not represent real observations.

These were treated as **structural missing data** rather than valid measurements.

To enable construction of complete aggregate time-series, a share-based backcasting method was applied:

1. A stable post-entry reference period was identified for each country.
2. Average national shares of total observed values were calculated.
3. These shares were applied retrospectively to estimate earlier missing values.

Estimated values were:

- Explicitly flagged in the dataset.
- Used exclusively for aggregate descriptive analysis.
- Never used for country-level inference.

4.6 Redistribution of "Unknown" Gender Records

Some injury records lacked gender identification.

To maintain analytical usability:

- “Unknown” cases were proportionally reassigned to Male/Female categories.
- Redistribution weights were derived from employment gender composition by country and year.
- Validation checks ensured proportional consistency after reassignment.

This approach preserved total counts while enabling gender-specific risk indicators.

4.7 Derived Analytical Indicators

To allow meaningful comparison across countries with different workforce sizes, normalized metrics were introduced.

Primary derived indicator:

Casualties per 1,000 Employees =
(Number of Recorded Casualties ÷ Employment Level) × 1,000

This normalization converts raw counts into comparable risk rates and forms the core analytical measure used throughout the database.

4.8 Output of the Preparation Phase

The preparation phase produced a curated CSV layer containing:

- Cleaned and harmonized injury datasets.
- Employment reference datasets.
- Derived incidence indicators.
- Explicit estimation flags for reconstructed observations.

These curated datasets served as the direct input for relational database construction described in Section 6.

4.9 Data Integrity Principles Applied

The preparation workflow followed three guiding principles:

1. **Preserve Observed Data**
   No transformation altered validated source values.
2. **Make Estimation Explicit**
   All reconstructed values are traceable and flagged.
3. **Avoid Artificial Continuity**
   Structural statistical breaks were respected rather than smoothed.

This ensured the analytical dataset remains statistically defensible and aligned with official data-production practices.

5. Treatment of Data Limitations

5.1 Overview

The source datasets originate from multiple EUROSTAT statistical domains and extraction views. As is common in multi-country administrative statistics, several comparability and completeness issues were identified during profiling.

Rather than masking these limitations through interpolation or smoothing, the curation strategy explicitly identified, documented, and treated each issue in a transparent manner consistent with official statistical practices.

5.2 Structural Breaks in Time-Series

Several national series contained periods with zero values that did not correspond to real-world observations but instead reflected:

- Delayed adoption of ESAW reporting frameworks.
- Changes in national data-collection methodology.
- Reclassification of accident-recording systems.
- Partial availability of circumstance-level reporting.

These values were interpreted as **structural missing data** rather than valid zero observations.

No interpolation was applied across these discontinuities to avoid introducing artificial trends.

5.3 Backcasting of Missing Historical Values

To support construction of coherent aggregate indicators across countries, a controlled estimation procedure was applied where national series entered the dataset later than others.

A proportional backcasting method was used:

1. A stable observation window following each country's data-entry point was identified.
2. The country's average share of the total population of recorded accidents was calculated within this window.
3. These shares were applied retrospectively to estimate earlier missing values in aggregate series.

This method is consistent with approaches used in official statistics when harmonizing datasets with staggered national participation.

Important safeguards:

- Estimated values are clearly flagged within the dataset.
- Backcasted values are used only for aggregated descriptive analysis.
- Country-level analytical results rely exclusively on observed data.

5.4 Interpretation of Zero Values in Residual Categories

In some datasets, residual classifications (e.g., “Other” categories) exhibited declining or zero values in later years.

These patterns were interpreted not as missing data but as improvements in classification specificity, where incidents were increasingly recorded under more detailed categories.

Therefore, no estimation or redistribution was performed for these cases.

5.5 Gender Information Gaps

Certain injury records contained the category “Unknown” for sex.

To preserve the analytical usefulness of gender-disaggregated indicators:

- Unknown cases were redistributed proportionally between male and female categories.
- Redistribution weights were derived from observed employment composition by country and year.
- Validation checks ensured that reassigned totals remained fully consistent with original aggregates.

This allowed computation of gender-specific incidence rates without altering total casualty counts.

5.6 Incompatibility of Age-Class Structures

A structural limitation exists between employment and injury datasets:

- Injury statistics use ESAW age bands.
- Employment statistics provide broader labor-force age classifications.

Because these classifications cannot be reconciled without introducing arbitrary assumptions, no age-standardized injury rates were calculated.

Age variables are therefore retained for descriptive purposes only and excluded from comparative rate analysis.

5.7 Differences in Dataset Dimensionality

Not all datasets include the same analytical attributes:

Dataset Type	Missing Attribute	Implication
Fatal / Non-fatal accident datasets	Age class	Age-specific fatality analysis not possible
Employment datasets	Detailed circumstance	Cannot model exposure by accident modality

Dataset Type	Missing Attribute	Implication
Injury datasets	Occupational structure	Limited linkage to ISCO-level risk

The database schema preserves these asymmetries while enabling valid comparisons across shared dimensions.

5.8 Analytical Scope Resulting from These Constraints

Given the identified limitations, the curated database is designed to support:

- Cross-country comparison of injury incidence rates.
- Within-country cross-activity comparison of injury incidence rates in Italy.
- Sector-level analysis of Transportation & Storage (NACE H).
- Circumstance-based pattern identification.
- Gender-disaggregated safety analysis.

The database is **not intended** for:

- Causal inference.
- Age-standardized epidemiological modeling.
- Country ranking based on estimated historical values.

5.9 Transparency and Traceability Measures

To ensure methodological transparency:

- All estimated observations include an explicit estimation flag.
- Original values remain recoverable from the curated layer.
- No transformation overwrites validated source data.
- All assumptions are documented in this section to support reproducibility.

This approach prioritizes statistical defensibility over artificial completeness.

6. Database Design

6.1 Design Objectives

Following data preparation, a relational analytical database was constructed to support consistent, multi-dimensional analysis of work-related injuries in the Transportation and Storage sector.

The database was designed to:

- Integrate injury statistics and employment data within a single analytical framework.
- Enable normalization of accident counts using workforce denominators.
- Support cross-country, within-country cross-activity, temporal, and categorical comparisons
- Preserve traceability to curated source datasets.
- Reduce redundancy while maintaining statistical transparency.

The resulting system is implemented as a SQLite database (WRI_1.0.db) optimized for reproducible statistical analysis rather than transactional use.

6.2 Schema Architecture

The database follows a **Multi-Star Schema** design.

This structure was selected because the project integrates multiple fact datasets - each with different analytical granularity - sharing a common set of descriptive dimensions.

The architecture consists of:

- Five fact tables containing quantitative observations.
- Seven dimension tables providing descriptive context.
- One-to-many relationships linking observations to shared classifications.

This design allows analysts to query different datasets independently or combine them through shared dimensions such as Country, Activity, Sex, and Year.

6.3 Analytical Grain

Each fact table was modeled at its natural statistical resolution to avoid artificial aggregation.

The most detailed dataset (WRI_CIRCUMSTANCE_HC10) follows the analytical grain:

Country × Year × Activity × Circumstance × Severity × Sex × Age

Other fact tables retain compatible but less granular structures (e.g., totals by activity or employment counts).

Maintaining native granularity ensures that no information is lost during integration.

6.4 Fact Tables

The database includes five core fact tables:

Fact Table	Role	Description
WRI_CIRCUMSTANCE_HC10	Detailed injury records	Accident data by circumstance for 10 EU countries
WRI_TOTALS_HC10	Aggregated injuries	Country-level totals and normalized indicators
WRI_IT_TOTALS_AU	Italy-focused dataset	Injuries by economic activity (A–U)
EMPLOYMENT_H36C	Exposure population	Employment counts for 36 countries (Activity H)
EMPLOYMENT_ITAU	National exposure	Employment by activity within Italy

These tables provide the measurable quantities used in analysis (casualties, employment levels, derived rates).

6.5 Dimension Tables

Seven dimension tables standardize classification systems extracted from the source data:

Dimension	Purpose
COUNTRIES	Geographic reporting entities
ACTIVITIES	NACE Rev. 2 economic sectors
SEX	Gender classification
AGE	Age-class groupings
SEVERITY	Injury severity categories
CIRCUMSTANCE_CATEGORIES	High-level accident context
CIRCUMSTANCE_SUBCATEGORIES	Detailed modality descriptions

This normalization eliminates duplication while ensuring consistency across datasets.

6.6 Data Standardization and Naming Conventions

To ensure clarity and interoperability:

- Table names use uppercase to distinguish entities.
- Column names use lowercase snake_case to support SQL readability.
- Identifier fields use *_id suffixes to denote relational keys.

- Descriptive metadata fields use *_description naming.

These conventions create a predictable structure for querying and future extension.

6.7 Referential Integrity

Primary and foreign keys enforce controlled relationships between fact and dimension tables.

Each observation in a fact table references valid classification records, ensuring:

- Consistent country and activity definitions across datasets.
- Elimination of orphan records.
- Reliable joins for rate calculations and comparative analysis.

This structure guarantees that all analytical outputs are dimensionally coherent.

6.8 Derived Measures Stored in the Database

Selected calculated indicators were retained within fact tables to ensure reproducibility of published results.

These include:

- Casualties per 1,000 employees.
- Gender-disaggregated casualty estimates.
- Employment-adjusted exposure metrics.
- Estimation flags identifying reconstructed observations.

Retaining these values allows analysts to reproduce results without re-performing preprocessing steps while still enabling recalculation if needed.

6.9 Data Reduction and Column Pruning

Columns not required for statistical interpretation—such as redundant labels, intermediate calculation fields, or Excel-specific artifacts—were removed prior to database loading.

This step reduced schema complexity and ensured that the database contains only analytically meaningful attributes.

6.10 Integration Logic

The database supports the integration of injury and employment datasets through shared dimensions rather than physical merging of source tables.

Normalized analytical indicators were initially computed during the data preparation phase (Excel-based transformation workflow) and stored in the database as calculated attributes. The relational structure allows these indicators to be recomputed through SQL joins between:

- Injury Counts (WRI tables)

and

- Employment Levels (EMPLOYMENT tables)

This dual approach enables verification of previously derived values while preserving the independence of the original statistical sources and ensuring methodological reproducibility.

6.11 Suitability of the Design for Statistical Analysis

The Multi-Star Schema provides several advantages for this type of occupational-safety analysis:

- Supports multi-level aggregation without redefining datasets.
- Maintains separation between observed and estimated values.
- Allows independent validation of injury and employment sources.
- Enables scalable extension to additional countries or years.
- Aligns with common statistical warehouse modeling practices.

The design therefore balances normalization, analytical flexibility, and methodological transparency.

6.12 Summary

The database layer transforms curated flat files into a structured analytical system that preserves statistical meaning while enabling reproducible querying.

The database functions as a structured analytical and validation environment, linking curated datasets and enabling reproducible querying and verification of indicators previously derived during the data-preparation phase.

7. Data Model

7.1 Conceptual Model

The curated database adopts a dimensional data-modeling approach designed to support statistical analysis rather than transactional processing.

The model separates **measurable observations (fact tables)** from **descriptive classifications (dimension tables)**, enabling flexible aggregation across multiple analytical perspectives.

This structure allows occupational injury data to be analyzed simultaneously by country, year, activity, demographic group, and accident circumstance.

7.2 Entity Relationships

The model consists of five fact tables connected to seven shared dimension tables through one-to-many relationships.

Dimension tables provide standardized reference values, while fact tables store quantitative measures such as casualty counts and employment levels.

This relational organization ensures that each observation can be traced to a unique combination of classifications without duplication of metadata.

7.3 Analytical Grain

Each fact table retains its natural level of detail to preserve statistical meaning.

The most granular table (WRI_CIRCUMSTANCE_HC10) follows the analytical grain:

Country × Year × Activity × Circumstance × Severity × Sex × Age

Other fact tables maintain compatible but less detailed grains (e.g., totals by activity or employment counts).

Maintaining native granularity prevents loss of information and supports both aggregated and disaggregated analysis.

7.4 Integration Logic

The model integrates injury and employment statistics through shared dimensional keys rather than physical merging.

While normalized indicators are stored as pre-calculated measures, the relational structure allows analysts to recompute these values through joins when validation or methodological transparency is required.

7.5 Advantages of the Adopted Model

The dimensional structure provides:

- Consistent aggregation across heterogeneous datasets.
- Reduced redundancy and improved clarity.
- Transparent linkage between numerator and denominator data.
- Scalability for additional years or countries.
- Compatibility with statistical and business-intelligence workflows.

The resulting data model supports reproducible and interpretable occupational-safety analysis.

8. Data Dictionary

8.1 Purpose

The data dictionary provides formal definitions of variables, classifications, and derived indicators included in the curated database.

It ensures consistent interpretation of fields and supports reproducibility of analytical results.

8.2 Identifier Fields

Identifier fields (suffix *_id*) represent standardized categorical keys used to join fact and dimension tables.

Examples include:

Field	Description
`country_id`	Reporting country code
`activity_id`	Economic activity (NACE Rev. 2)
`sex_id`	Gender classification
`age_id`	Age class
`severity_id`	Injury severity category
`subcategory_id`	Detailed accident circumstance

These identifiers enforce referential integrity and prevent duplication of descriptive labels.

8.3 Measure Fields

Measure fields store quantitative observations derived from EUROSTAT datasets.

Field	Meaning
`obs_value`	Recorded number of injuries or employees
`employment_h / employment_au`	Workforce exposure (thousands of persons)
`casualties_x1k`	Injuries per 1,000 employees

Field	Meaning
`m_casualties / f_casualties`	Gender-disaggregated casualty estimates

All measures retain their original statistical interpretation unless explicitly marked as derived.

8.4 Derived Variables

Derived indicators were calculated to support comparability:

- Casualties per 1,000 employees (normalized risk measure).
- Gender-adjusted counts for records with missing sex.
- Estimation flags identifying reconstructed observations.

Derived fields remain traceable to their source variables.

8.5 Metadata and Descriptive Fields

Descriptive attributes (suffix *_description*) provide human-readable labels associated with identifiers.

These fields improve interpretability without altering the underlying categorical structure.

8.6 Naming Convention

The following naming rules were applied:

- Lowercase snake_case for column names.
- Uppercase table names.
- *_id* suffix for keys.
- *_description* suffix for labels.
- Numeric fields expressed using standard SI units where applicable.

This convention ensures clarity, consistency, and SQL compatibility.

9. Reproducibility

9.1 Reproducibility Objective

The curation workflow was designed to allow reconstruction of the analytical dataset from publicly available EUROSTAT extractions using deterministic transformation steps.

9.2 Processing Environment

The workflow was implemented using:

• CSV data extracted from EUROSTAT.
• Spreadsheet-based profiling and transformation.
• SQLite database environment for relational modeling.

No proprietary or opaque transformation processes were applied.

9.3 Transformation Traceability

Each stage of the pipeline produces an auditable intermediate output:

Stage	Output
Raw extraction	Source CSV files
Data preparation	Cleaned and harmonized CSV layer
Database loading	Structured relational schema
Analytical layer	Derived indicators and joins

All derived values can be recalculated using documented formulas.

The original calculations were executed during spreadsheet-based preparation and are reproducible within the database using equivalent relational queries.

9.4 Version Control and Transparency

- Source datasets remain unchanged and recoverable.
- Estimated values are explicitly flagged.
- Schema design preserves links to original statistical variables.

This ensures analytical conclusions can be validated independently.

9.5 Reuse Potential

Because the workflow relies on standardized classifications and open data, the database can be extended to include:

- Additional years of EUROSTAT releases.
- Other NACE sectors.
- Expanded geographic coverage.

The methodology is therefore reusable beyond the scope of this study.

9.6 Use of AI-Assisted Tools

Artificial Intelligence tools were used solely to support editorial and documentation activities.

All substantive analytical decisions were independently defined by the author prior to any AI use.

Specifically, the author solely determined:

- Selection of the research topic and analytical scope.
- Identification of relevant statistical domains (ESAW and LFS).
- Choice of datasets, filters, and variables.
- Definition of comparison logic (cross-country within Activity H; cross-sector within Italy).
- Data-preparation methodology, including estimation strategy and derived indicators.
- Design of the relational database schema and analytical grain.
- Implementation of transformations, calculations, and validation procedures.

AI tools were subsequently used only to assist with:

- Language editing (grammar, clarity, and terminology standardization).
- Structuring descriptive documentation of already-implemented workflows.
- Formatting tables and summaries for readability.
- Producing visual representations (e.g., ERD diagram) based strictly on the predefined schema.
- Verifying consistency between written explanations and the executed methodology.
- Finding a suitable structural template to improve clarity and consistency.

10. Limitations

10.1 Statistical Constraints

The curated dataset inherits limitations from the underlying administrative statistics, including:

- Differences in national reporting practices.
- Structural breaks in ESAW time-series.
- Incomplete demographic coverage in some datasets.

These constraints were documented rather than statistically imputed.

10.2 Comparability Restrictions

Age classifications differ between employment and injury datasets, preventing construction of age-standardized risk indicators.

Similarly, not all datasets contain identical dimensional variables, limiting certain cross-tabulations.

10.3 Interpretation of Estimated Values

Backcasted observations were introduced only to maintain aggregate continuity and must not be interpreted as observed historical data.

Analytical conclusions should therefore prioritize reported values over reconstructed ones.

10.4 Analytical Scope

The dataset supports descriptive and comparative analysis but is not intended for:

- Causal inference.
- Predictive modeling.
- Epidemiological standardization.
- Regulatory benchmarking.

10.5 Dependence on Source Updates

Future EUROSTAT revisions may alter classifications or historical values, requiring updates to maintain alignment with official statistics.

Appendices

Appendix A – EUROSTAT Dataset Inventory

The following tables list all EUROSTAT datasets extracted for this project, including their statistical purpose, applied filters, and analytical role in the curated database.

These files constitute the full set of external inputs used during the Excel-based preparation phase prior to relational database construction.

A.1 ESAW Detailed Accident Datasets (Circumstance-Level Sources)

These datasets provide multidimensional ESAW records describing the circumstances of occupational accidents.

They were harmonized and consolidated during Phase 1 to construct the detailed analytical table used for circumstance-based analysis.

Dataset A1 — Workstation Dimension

Field	Value
File Name	hsw_ph3_01__custom_19447496_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and workstation
Analytical Role	Circumstance analysis – Workstation
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	“Total” removed for Sex, Age, Workstation
Records	9,600

Field	Value
Variables	27 columns
File Size	2.6 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A2 — Working Environment Dimension

Field	Value
File Name	hsw_ph3_02__custom_19447534_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and working environment
Analytical Role	Circumstance analysis – Working Environment
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	“Total” removed for Sex, Age, Working Environment
Records	33,600
Variables	27 columns
File Size	9.4 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as

Field	Value
	WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A3 — Working Process Dimension

Field	Value
File Name	hsw_ph3_03__custom_19447620_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and working process
Analytical Role	Circumstance analysis – Working Process
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	“Total” removed for Sex, Age, Working Process
Records	19,200
Variables	27 columns
File Size	6.0 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A4 — Specific Physical Activity Dimension

Field	Value
File Name	hsw_ph3_04__custom_19447665_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and specific physical activity
Analytical Role	Circumstance analysis – Specific Physical Activity
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	"Total" removed for Sex, Age, Specific Physical Activity
Records	19,200
Variables	27 columns
File Size	5.3 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A5 — Deviation Dimension

Field	Value
File Name	hsw_ph3_06__custom_19447710_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and deviation

Field	Value
Analytical Role	Circumstance analysis – Deviation (Triggering Event)
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	"Total" removed for Sex, Age, Deviation
Records	21,600
Variables	27 columns
File Size	6.6 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A6 — Contact Mode of Injury Dimension

Field	Value
File Name	hsw_ph3_08__custom_19447748_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Accidents at work by sex, age, severity, activity and contact mode of injury
Analytical Role	Circumstance analysis – Contact Mode of Injury
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H

Field	Value
Time Span	2014–2023
Exclusions	“Total” removed for Sex, Age, Contact Mode
Records	21,600
Variables	27 columns
File Size	6.7 MB
Transformation Lineage	Records were integrated with the other ESAW circumstance extracts (A1–A6) into a single analytical worksheet during Phase 1. The curated dataset was exported as WRI_C10_H_CIRCUMSTANCE.csv and subsequently ingested into SQLite as WRI_CIRCUMSTANCE_HC10.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

A.2 ESAW Aggregated Datasets (Totals and Validation)

These datasets provide aggregated counts of fatal and non-fatal accidents at work and were used to construct validated national totals and normalized indicators.

They represent event-level statistics and are not directly comparable with the detailed circumstance datasets in Section A.1, where a single accident may be described across multiple ESAW classification dimensions.

For this reason, circumstance-level frequencies should be interpreted as distributions of accident characteristics rather than as additive components of total accident counts.

Dataset A7 — Non-Fatal Accidents (10 Countries)

Field	Value
File Name	hsw_n2_01__custom_19447306_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Non-fatal accidents at work by NACE Rev.2 activity and sex
Analytical Role	Aggregate injury totals (Non-fatal component)
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Exclusions	None (Sex total retained for aggregation)
Records	100
Variables	21 columns
File Size	18 KB
Transformation Lineage	Records were consolidated with the complementary ESAW totals extract (A7–A8) during Phase 1 (Excel-based preparation). The curated dataset was exported as WRI_C10_TOT_H.csv and subsequently ingested into SQLite as WRI_TOTALS_HC10 for Phase 2 database construction.

Field	Value
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A8 — Fatal Accidents (10 Countries)

Field	Value
File Name	hsw_n2_02__custom_19447318_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Fatal accidents at work by NACE Rev.2 activity
Analytical Role	Aggregate injury totals (Fatal component)
Countries	Germany, France, Italy, Spain, Poland, Romania, Netherlands, Czechia, Hungary, Belgium
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Records	100
Variables	19 columns
File Size	16 KB
Transformation Lineage	Records were consolidated with the complementary ESAW totals extract (A7–A8) during Phase 1 (Excel-based preparation). The curated dataset was exported as WRI_C10_TOT_H.csv and subsequently ingested into SQLite as WRI_TOTALS_HC10 for Phase 2 database construction.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

A.3 Employment Datasets Used as Exposure Denominators (LFS Domain)

These datasets provide the employment reference population required to normalize injury counts.

They originate from the EUROSTAT Labour Force Survey and were prepared during Phase 1 to supply denominator values for incidence-rate calculations at both the European and national levels.

Dataset A9 — Employment by Activity H (36 Countries)

Field	Value
File Name	lfsa_eisn2__custom_19447243_linear_2_0.csv
Source	EUROSTAT – Labour Force Survey (LFS)
Description	Employed persons by occupation and economic activity (NACE Rev.2)
Analytical Role	Exposure denominator for incidence-rate calculations
Countries	36 European countries
Activity Filter	NACE Rev.2 – Activity H
Time Span	2014–2023
Records	346
Variables	25 columns
File Size	85 KB
Transformation Lineage	The dataset was imported into Excel for Phase 1 profiling and harmonization. The curated output was exported as Employment_H_36_C.csv and subsequently ingested into SQLite as EMPLOYMENT_H36C, providing the reference employment population used to compute normalized indicators.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A10 — Employment by Activity (Italy, NACE A–U)

Field	Value
File Name	lfsa_eisn2__custom_19484136_linear_2_0.csv

Field	Value
Source	EUROSTAT – Labour Force Survey (LFS)
Description	Employed persons by economic activity (NACE Rev.2, A–U)
Analytical Role	Exposure denominator for within-country sector comparison
Country	Italy
Activity Coverage	NACE Rev.2 Sections A–U
Time Span	2014–2023
Records	210
Variables	25 columns
File Size	55 KB
Transformation Lineage	The dataset was imported into Excel for Phase 1 profiling and harmonization. The curated output was exported as Employment_IT_AU.csv and subsequently ingested into SQLite as EMPLOYMENT_ITAU, to support sector-level normalization within Italy.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

A.4 Italy ESAW Datasets for Cross-Activity Comparison (NACE A–U)

These datasets provide aggregated accident counts for Italy across NACE Rev.2 activities (A–U) and were harmonized to support within-country sectoral comparison.

The extracts represent event-level statistics and were consolidated during Phase 1 into a single curated dataset prior to database integration.

Dataset A11 — Non-Fatal Accidents by Activity (Italy, A–U)

Field	Value
File Name	hsw_n2_01__custom_19448164_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Non-fatal accidents at work by economic activity (NACE Rev.2)
Analytical Role	Component of Italy cross-activity injury totals
Country	Italy
Activity Coverage	NACE Rev.2 Sections A–U
Time Span	2014–2023
Records	210
Variables	21 columns
File Size	40 KB
Transformation Lineage	Records were consolidated with the complementary ESAW extract (A11–A12) during Phase 1 (Excel-based preparation). The curated dataset was exported as WRI_IT_TOT_AU.csv and subsequently ingested into SQLite as WRI_IT_TOTALS_AU for Phase 2 database construction.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Dataset A12 — Fatal Accidents by Activity (Italy, A–U)

Field	Value
File Name	hsw_n2_02__custom_19448203_linear_2_0.csv
Source	EUROSTAT – ESAW
Description	Fatal accidents at work by economic activity (NACE Rev.2)
Analytical Role	Component of Italy cross-activity injury totals
Country	Italy
Activity Coverage	NACE Rev.2 Sections A–U
Time Span	2014–2023
Records	210
Variables	19 columns
File Size	35 KB
Transformation Lineage	Records were consolidated with the complementary ESAW extract (A11–A12) during Phase 1 (Excel-based preparation). The curated dataset was exported as WRI_IT_TOT_AU.csv and subsequently ingested into SQLite as WRI_IT_TOTALS_AU for Phase 2 database construction.
EUROSTAT Access	https://ec.europa.eu/eurostat/databrowser

Appendix B – Phase 1 Outputs (Curated CSV Layer)

This appendix documents the structure of the curated CSV datasets produced during Phase 1 (Excel-based preparation). These files represent the harmonized analytical layer derived from EUROSTAT extracts prior to loading into the SQLite database.

Table B1. Structure of `WRI_C10_H_CIRCUMSTANCE.csv` used to populate `WRI_CIRCUMSTANCE_HC10`

Column Name	Type	Description
STRUCTURE	STRING	The data origin: specifies whether the record derives from dataflow or backcasting.
STRUCTURE_ID	STRING	Data origin identifier.
STRUCTURE_NAME	STRING	Category of circumstance description.
freq	STRING	Time Frequency ID.
Time frequency	STRING	Time Frequency description.
wrkstat	STRING	Circumstance ID.
Circumstance	STRING	Circumstance description.
severity	STRING	Severity ID.
Severity (days lost)	STRING	Severity description (days lost classification).
nace_r2	STRING	NACE Rev. 2 ID.
Statistical classification of economic activities in the European Community (NACE Rev. 2)	STRING	NACE Rev. 2 description.
sex	STRING	Gender ID.
Sex	STRING	Gender description.
age	STRING	Age class ID.
Age class	STRING	Age class description.

Column Name	Type	Description
unit	STRING	Unit of measure ID.
Unit of measure	STRING	Unit of measure description.
geo	STRING	Geopolitical entity ID (reporting country).
Geopolitical entity (reporting)	STRING	Geopolitical entity description.
TIME_PERIOD	NUMBER	Year in which the accident was recorded.
OBS_VALUE	NUMBER	Number of casualties recorded.
OBS_FLAG	STRING	Observation status ID.
Observation status (Flag) V2 structure	STRING	Observation status description (e.g., break in time series).
Employment H	NUMBER	Employees (Thousand persons).
Casualties per 1k employees	NUMBER	Number of casualties per one thousand employees.
Estimation	BOOLEAN	TRUE = estimated; FALSE = original imported record.
% Males/year	PERCENTAGE	Percentage of male employees per year.
Male weight	NUMBER	Male employment assigned weight.
Female weight	NUMBER	Female employment assigned weight.
Weights check	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
% Male/Casualties	PERCENTAGE	Percentage of male casualties.
Casualties M weight	NUMBER	Male casualties assigned weight.
Casualties F weight	NUMBER	Female casualties assigned weight.
Weight check 2	BOOLEAN	PASSED = correct assignment; FAILED = incorrect

Column Name	Type	Description
		assignment.
Casualties M	NUMBER	Casualties assigned to the male gender.
Casualties F	NUMBER	Casualties assigned to the female gender.
Employment H M	NUMBER	Male employees (thousand persons) assigned to the male gender.
Employment H F	NUMBER	Female employees (thousand persons) assigned to the female gender.
Check	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Casualties M/1K M	NUMBER	Number of male casualties per 1,000 male employees.
Casualties F/1K F	NUMBER	Number of female casualties per 1,000 female employees.

Dataset contains 124,897 rows and 41 columns. CSV file size: 37,691 KB.

Table B2. Structure of WRI_C10_TOT_H.csv used to populate WRI_TOTALS_HC10

Column Name	Type	Description
STRUCTURE	STRING	The data origin: specifies whether the record derives from dataflow or backcasting.
STRUCTURE_ID	STRING	Data origin identifier.
STRUCTURE_NAME	STRING	Structure description.
freq	STRING	Time Frequency ID.
Time frequency	STRING	Time Frequency description.
unit	STRING	Unit of measure ID.
Unit of measure	STRING	Unit of measure description.

Column Name	Type	Description
nace_r2	STRING	NACE Rev. 2 ID.
Statistical classification of economic activities in the European Community (NACE Rev. 2)	STRING	NACE Rev. 2 description.
sex	STRING	Gender ID.
Sex	STRING	Gender description.
geo	STRING	Geopolitical entity ID (reporting country).
Geopolitical entity (reporting)	STRING	Geopolitical entity description.
TIME_PERIOD	NUMBER	Year in which the accident was recorded.
OBS_VALUE	NUMBER	Number of casualties recorded.
OBS_FLAG	STRING	Observation status ID.
Observation status (Flag) V2 structure	STRING	Observation status description (e.g., break in time series).
Employment H	NUMBER	Employees (Thousand persons).
Casualties per 1k employees	NUMBER	Number of casualties per one thousand employees.
% F/year	PERCENTAGE	Percentage of female employees per year.
M weight/y	NUMBER	Male employment assigned weight.
F weight/Y	NUMBER	Female employment assigned weight.
Check 1	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Emp M	NUMBER	Male employees (thousand persons) assigned to the male gender.
Emp F	NUMBER	Female employees (thousand persons) assigned to

Column Name	Type	Description
		the female gender.
Check 2	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
% Cas F	PERCENTAGE	Percentage of female casualties.
Cas M Weight	NUMBER	Male casualties assigned weight.
Cas F Weight	NUMBER	Female casualties assigned weight.
Check 3	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Cas M	NUMBER	Casualties assigned to the male gender.
Cas F	NUMBER	Casualties assigned to the female gender.
Check 4	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Cas M / 1k	NUMBER	Number of male casualties per 1,000 male employees.
Cas F / 1k	NUMBER	Number of female casualties per 1,000 female employees.

Dataset contains 400 rows and 35 columns. CSV file size: 87 KB.

Table B3. Structure of WRI_IT_TOT_AU.csv used to populate WRI_IT_TOTALS_AU.

Column Name	Type	Description
STRUCTURE	STRING	The data origin: specifies whether the record derives from dataflow or backcasting.
STRUCTURE_ID	STRING	Data origin identifier.
STRUCTURE_NAME	STRING	Structure description.
freq	STRING	Time Frequency ID.

Column Name	Type	Description
Time frequency	STRING	Time Frequency description.
unit	STRING	Unit of measure ID.
Unit of measure	STRING	Unit of measure description.
nace_r2	STRING	NACE Rev. 2 ID.
Statistical classification of economic activities in the European Community (NACE Rev. 2)	STRING	NACE Rev. 2 description.
sex	STRING	Gender ID.
Sex	STRING	Gender description.
geo	STRING	Geopolitical entity ID (reporting country).
Geopolitical entity (reporting)	STRING	Geopolitical entity description.
TIME_PERIOD	NUMBER	Year in which the accident was recorded.
OBS_VALUE	NUMBER	Number of casualties recorded.
OBS_FLAG	STRING	Observation status ID.
Observation status (Flag) V2 structure	STRING	Observation status description (e.g., break in time series).
Emplyment AU	NUMBER	Employees (Thousand persons).
Casualties per 1k employees	NUMBER	Number of casualties per one thousand employees.
% Males/Year	PERCENTAGE	Percentage of male employees per year.
M Weight	NUMBER	Male employment assigned weight per year.
F Weight	NUMBER	Female employment assigned weight per year.

Column Name	Type	Description
Check W	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
% F Cas./Y	PERCENTAGE	Percentage of female casualties per year.
Cas M Weight	NUMBER	Male casualties assigned weight.
Cas F Weight	NUMBER	Female casualties assigned weight.
Check 2	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Casualties M	NUMBER	Casualties assigned to the male gender.
Casualties F	NUMBER	Casualties assigned to the female gender.
Check 3	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Emp AU M	NUMBER	Male employees (thousand persons) assigned to the male gender.
Emp AU F	NUMBER	Female employees (thousand persons) assigned to the female gender.
Check 4	BOOLEAN	PASSED = correct assignment; FAILED = incorrect assignment.
Cas M/1K	NUMBER	Number of male casualties per 1,000 male employees.
Cas F/1K	NUMBER	Number of female casualties per 1,000 female employees.

Dataset contains 840 rows and 35 columns. CSV file size: 190 KB.

Table B4. Structure of Employment_H_36_C.csv used to populate EMPLOYMENT_H36C.

Column Name	Type	Description
STRUCTURE	STRING	The data origin: specifies whether the record derives from dataflow or backcasting.

STRUCTURE_ID	STRING	Data origin identifier.
STRUCTURE_NAME	STRING	Structure description.
freq	STRING	Time Frequency ID.
Time frequency	STRING	Time Frequency description.
age	STRING	Age class ID.
Age class	STRING	Age class description.
sex	STRING	Gender ID.
Sex	STRING	Gender description.
nace_r2	STRING	NACE Rev. 2 ID.
Statistical classification of economic activities in the European Community (NACE Rev. 2)	STRING	NACE Rev. 2 description.
isco08	STRING	ISCO-08 ID.
International Standard Classification of Occupations 2008 (ISCO-08)	STRING	ISCO-08 description.
unit	STRING	Unit of measure ID.
Unit of measure	STRING	Unit of measure description.
geo	STRING	Geopolitical entity ID (reporting country).
Geopolitical entity (reporting)	STRING	Geopolitical entity description.
TIME_PERIOD	NUMBER	Year in which the datum was recorded.
OBS_VALUE	NUMBER	Number of employees recorded.
OBS_FLAG	STRING	Observation status ID.

Observation status (Flag) V2 structure	STRING	Observation status description (e.g., break in time series).

Dataset contains 692 rows and 21 columns. CSV file size: 167 KB.

Table B5. Structure of Employment_IT_AU.csv used to populate EMPLOYMENT_ITAU.

Column Name	Type	Description
STRUCTURE	STRING	The data origin: specifies whether the record derives from dataflow or backcasting.
STRUCTURE_ID	STRING	Data origin identifier.
STRUCTURE_NAME	STRING	Structure description.
freq	STRING	Time Frequency ID.
Time frequency	STRING	Time Frequency description.
age	STRING	Age class ID.
Age class	STRING	Age class description.
sex	STRING	Gender ID.
Sex	STRING	Gender description.
nace_r2	STRING	NACE Rev. 2 ID.
Statistical classification of economic activities in the European Community (NACE Rev. 2)	STRING	NACE Rev. 2 description.
isco08	STRING	ISCO-08 ID.
International Standard Classification of Occupations 2008 (ISCO-08)	STRING	ISCO-08 description.
unit	STRING	Unit of measure ID.
Unit of measure	STRING	Unit of measure description.

geo	STRING	Geopolitical entity ID (reporting country).
Geopolitical entity (reporting)	STRING	Geopolitical entity description.
TIME_PERIOD	NUMBER	Year in which the datum was recorded.
OBS_VALUE	NUMBER	Number of employees recorded.
OBS_FLAG	STRING	Observation status ID.
Observation status (Flag) V2 structure	STRING	Observation status description (e.g., break in time series).

Dataset contains 420 rows and 21 columns. CSV file size: 108 KB.

Fields marked as calculated (e.g., normalized rates or gender redistributions) were derived within Excel and stored explicitly to ensure auditability prior to database normalization.

Appendix C – Derived Indicator Formulae

Casualties per 1,000 Employees =
(Recorded Casualties ÷ Employment Level) × 1,000

When employment data are expressed in thousands of persons (as in EUROSTAT LFS extracts), the indicator can be computed directly as Recorded Casualties divided by Employment, since the scaling factor of 1,000 is already embedded in the unit of measurement.

Gender Redistribution =
Unknown Cases × Employment Gender Share

Appendix D – Phase 2 Outputs (SQLite Database Layer)

Phase 2 reorganizes the curated CSV datasets into a relational multi-star schema. No new indicators are calculated within SQLite; the database stores and structures the curated values to enable reproducible querying and validation.

D1. Overview

Table D1.1 Dimension Tables Overview

All dimension tables contain categorical lookups only.

Table Name	Purpose	Key Field
COUNTRIES	Geographic reference	country_id
ACTIVITIES	NACE classification	activity_id
AGE	Age categories	age_id
SEX	Gender categories	sex_id
SEVERITY	Injury severity	severity_id
CIRCUMSTANCE_CATEGORIES	Accident context groups	category_id
CIRCUMSTANCE_SUBCATEGORIES	Detailed modalities	subcategory_id

Table D1.2. Fact Tables Overview

Table Name	Source CSV	Analytical Role
WRI_CIRCUMSTANCE_HC10	WRI_C10_H_CIRCUMSTANCE	Detailed circumstance analysis
WRI_TOTALS_HC10	WRI_C10_TOT_H	Aggregated country totals
WRI_IT_TOTALS_AU	WRI_IT_TOT_AU	Italy cross-sector analysis
EMPLOYMENT_H36C	Employment_H_36_C	Exposure denominator
EMPLOYMENT_ITAU	Employment_IT_AU	National exposure

D2. Dimension tables

Table D2.1. Structure of table COUNTRIES

Reference table containing the geographic identifiers used to link employment and injury datasets across all fact tables.

Column Name	Type	Key	Description
country_id	TEXT	PK	Country code
country	TEXT	–	Country name (36 values)

Dataset contains 36 rows and 2 columns.

Table D2.2. Structure of table ACTIVITIES

Lookup table for NACE Rev.2 economic activities used to classify observations by sector.

Column Name	Type	Key	Description
activity_id	TEXT	PK	NACE Rev.2 activity code
activity_description	TEXT	–	Activity label (21 unique values)

Dataset contains 21 rows and 2 columns.

Table D2.3. Structure of table AGE

Dimension table defining harmonized age classes adopted in ESAW and Labour Force Survey extracts.

Column Name	Type	Key	Description
age_id	TEXT	PK	Age class code
age_class_description	TEXT	–	Age category label (5 values)

Dataset contains 5 rows and 2 columns.

Table D2.4. Structure of table SEVERITY

Classification of injury outcomes distinguishing fatal and non-fatal events.

Column Name	Type	Key	Description
severity_id	TEXT	PK	Severity code
severity_description	TEXT	–	Fatal / Non-fatal classification

Dataset contains 2 rows and 2 columns.

Table D2.5. Structure of table SEX

Standardized gender classification used to align ESAW and employment data and to support redistributed indicators.

Column Name	Type	Key	Description
sex_id	TEXT	PK	Sex category code
sex_description	TEXT	–	Male / Female / Unknown

Dataset contains 3 rows and 2 columns.

Table D2.6. Structure of table CIRCUMSTANCE_CATEGORIES

High-level ESAW taxonomy describing the general context in which accidents occurred.

Column Name	Type	Key	Description
category_id	TEXT	PK	ESAW category identifier
category_description	TEXT	–	Circumstance group (6 values)

Dataset contains 6 rows and 2 columns.

Table D2.7. Structure of table CIRCUMSTANCE_SUBCATEGORIES

Detailed ESAW classification expanding circumstance categories into specific modalities used for analytical disaggregation.

Column Name	Type	Key	Description
subcategory_id	TEXT	PK	ESAW subcategory code
category_id	TEXT	FK	Links to CIRCUMSTANCE_CATEGORIES
subcategory_description	TEXT	–	Detailed modality (52 values)

Dataset contains 52 rows and 3 columns.

D3. Fact tables

Table D3.1. Structure of table EMPLOYMENT_H36C

Employment dataset covering the 36-country reference population used as the denominator for cross-country incidence calculations.

Column Name	Type	Key	Description
eh_id	INTEGER	PK	Record identifier
structure	TEXT	–	Data structure type
structure_name	TEXT	–	Structure description
time_frequency	TEXT	–	Time aggregation
age_id	TEXT	FK	Age reference
sex_id	TEXT	FK	Sex reference
activity_id	TEXT	FK	Activity reference
isco08_description	TEXT	–	Occupation description
unit_description	TEXT	–	Unit metadata
country_id	TEXT	FK	Country reference
time_period	INTEGER	–	Year
obs_value	REAL	–	Employees (thousand persons)
observation_status	TEXT	–	Data status flag

Dataset contains 692 rows and 13 columns.

Table D3.2. Structure of table EMPLOYMENT_ITAU

Italy-specific employment dataset used to compute sector-level indicators across NACE A–U activities.

Column Name	Type	Key	Description
eit_id	INTEGER	PK	Record identifier

Column Name	Type	Key	Description
structure	TEXT	–	Structure type
structure_name	TEXT	–	Structure description
time_frequency	TEXT	–	Time aggregation
age_id	TEXT	FK	Age reference
sex_id	TEXT	FK	Sex reference
activity_id	TEXT	FK	Activity reference
isco08_description	TEXT	–	Occupation description
unit_description	TEXT	–	Unit metadata
country_id	TEXT	FK	Country reference
time_period	INTEGER	–	Year
obs_value	REAL	–	Employees (thousand persons)
observation_status	TEXT	–	Data status flag

Dataset contains 420 rows and 13 columns.

Table D3.3. Structure of table WRI_CIRCUMSTANCE_HC10

Detailed analytical table combining ESAW circumstance descriptors with employment exposure to enable multidimensional injury analysis.

Column Name	Type	Key	Description
wri_id	INTEGER	PK	Record identifier
structure	TEXT	–	Structure type
category_id	TEXT	FK	Circumstance category
time_frequency	TEXT	–	Time frequency description

Column Name	Type	Key	Description
subcategory_id	TEXT	FK	Circumstance detail
severity_id	TEXT	FK	Severity reference
activity_id	TEXT	FK	Activity reference
sex_id	TEXT	FK	Sex reference
age_id	TEXT	FK	Age reference
unit_description	TEXT	–	Unit of measure description
country_id	TEXT	FK	Country reference
time_period	INTEGER	–	Year
obs_value	REAL	–	Observed value (number of casualties)
observation_status	TEXT	–	Observation status description
employment_h	REAL	–	Employees (thousand persons). Source: Dataflow, imported into Excel
casualties_x1k	REAL	–	Casualties per thousand persons. Source: Calculated field in Excel
estimation_tot_gen	INTEGER	–	Redistribution flag
m_casualties	REAL	–	Male casualties. Source: Calculated field in Excel
f_casualties	REAL	–	Female casualties. Source: Calculated field in Excel
m_casualties_x1km	REAL	–	Casualties per thousand males. Source: Calculated field in Excel
f_casualties_x1kf	REAL	–	Casualties per thousand females. Source: Calculated field in Excel

Dataset contains 124,897 rows and 21 columns

Table D3.4. Structure of table WRI_IT_TOTALS_AU

Aggregated Italian dataset supporting within-country comparison of injury incidence across economic activities.

Column Name	Type	Key	Description
wri_it_id	INTEGER	PK	Record identifier
structure	TEXT	–	Structure type
severity_id	TEXT	FK	Severity reference
time_frequency	TEXT	–	Time frequency description
unit_description	TEXT	–	Unit of measure description
activity_id	TEXT	FK	Activity reference
sex_id	TEXT	FK	Sex reference
country_id	TEXT	FK	Country reference
time_period	INTEGER	–	Year
obs_value	REAL	–	Observed value (number of casualties)
observation_status	TEXT	–	Observation status description
employment_au	REAL	–	Employees (thousand persons). Source: Dataflow, imported into Excel
casualties_x1k	REAL	–	Casualties per thousand persons. Source: Calculated field in Excel
m_casualties	REAL	–	Male casualties. Source: Calculated field in Excel
f_casualties	REAL	–	Female casualties. Source: Calculated field in Excel
m_employment_au	REAL	–	Male employees. Unknown sex proportionally redistributed
f_employment_au	REAL	–	Female employees. Unknown sex

Column Name	Type	Key	Description
			proportionally redistributed
m_casualties_x1km	REAL	–	Casualties per thousand males. Source: Calculated field in Excel
f_casualties_x1kf	REAL	–	Casualties per thousand females. Source: Calculated field in Excel

Dataset contains 840 rows and 19 columns.

Table D3.5. Structure of table WRI_TOTALS_HC10

Aggregated cross-country dataset providing validated totals and normalized indicators for the HC10 comparison group.

Column Name	Type	Key	Description
wri_tot_id	INTEGER	PK	Record identifier
structure	TEXT	–	Structure type
severity_id	TEXT	FK	Severity reference
time_frequency	TEXT	–	Time frequency description
unit_description	TEXT	–	Unit of measure description
activity_id	TEXT	FK	Activity reference
sex_id	TEXT	FK	Sex reference
country_id	TEXT	FK	Country reference
time_period	INTEGER	–	Year
obs_value	REAL	–	Observed value (number of casualties)
observation_status	TEXT	–	Observation status description
employment_h	REAL	–	Employees (thousands of persons). Source: Dataflow, imported into Excel

Column Name	Type	Key	Description
casualties_x1k	REAL	–	Casualties per thousand persons. Source: Calculated field in Excel
m_employment_h	REAL	–	Male employees. Unknown sex proportionally assigned to males and females; known sex requires no estimation
f_employment_h	REAL	–	Female employees. Unknown sex proportionally assigned to males and females; known sex requires no estimation
m_casualties	REAL	–	Male casualties. Source: Calculated field in Excel
f_casualties	REAL	–	Female casualties. Source: Calculated field in Excel
m_casualties_x1km	REAL	–	Casualties per thousand males. Source: Calculated field in Excel
f_casualties_x1kf	REAL	–	Casualties per thousand females. Source: Calculated field in Excel

Dataset contains 400 rows and 21 columns

Derived indicators included in the fact tables (e.g., casualties per 1,000 employees and gender-redistributed values) were calculated during Phase 1 preparation and stored to preserve methodological traceability.

D4. Data Lineage

Table D4.1. Data lineage

The following table summarizes the transformation pathway from the Excel-based preparation layer (Phase 1) to the SQLite analytical schema (Phase 2).

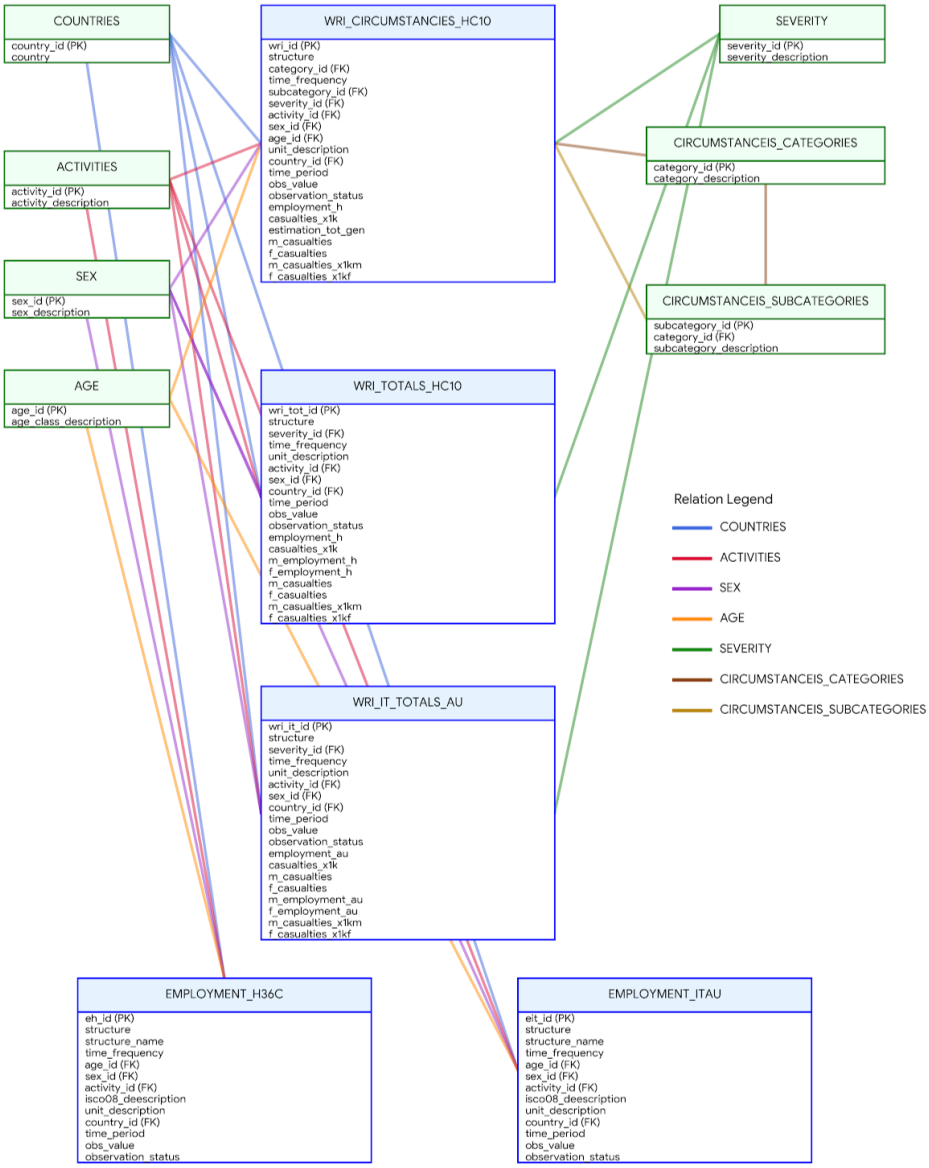
Phase	Output	Used To Create	Purpose
Excel Preparation	WRI_C10_H_CIRCUMSTANCE.csv	WRI_CIRCUMSTANCE_HC10	Circumstance analysis
Excel Preparation	WRI_C10_TOT_H.csv	WRI_TOTALS_HC10	Aggregated validation totals
Excel Preparation	WRI_IT_TOT_AU.csv	WRI_IT_TOTALS_AU	Italy cross-activity comparison
Excel Preparation	Employment_H_36_C.csv	EMPLOYMENT_H36C	Exposure denominator
Excel Preparation	Employment_IT_AU.csv	EMPLOYMENT_ITAU	National employment reference

Unlike Phase 1, no new indicators were calculated within SQLite. The database layer preserves and organizes the curated values while enabling relational validation and multidimensional analysis.

Appendix E – Entity–Relationship Diagram (ERD)

Figure E1.1. Entity–Relationship Diagram (ERD) of the implemented multi-star schema architecture.

The diagram represents the relational architecture implemented in SQLite.



Dimension tables (green) provide ESAW and Labour Force Survey classification attributes, while fact tables (blue) store curated injury and employment measures produced during Phase 1 preparation.

The vertical arrangement reflects the analytical workflow: classification dimensions → detailed circumstance records → aggregated injury indicators → employment exposure datasets.

Appendix F – Classification Systems Used

The analysis integrates multiple European statistical classifications to ensure comparability between injury records and employment data. The following systems were applied:

- NACE Rev. 2 – Classification of economic activities used to identify sectoral scope (e.g., Activity H and A–U comparisons).
- ESAW Variables – European Statistics on Accidents at Work classifications describing accident circumstances, severity, and demographic characteristics.
- ISCO-08 – International Standard Classification of Occupations used in the Labour Force Survey employment datasets.

Appendix G – Excel Transformations (Pivot & Formula Schema)

G.1 Pivot Tables (Aggregation Layer)

1. Pivot Emp – Employment (36 Countries, Activity H)

- **Source**: Employment H 36 C
- **Rows**: Country (geo)
- **Columns**: Year (TIME_PERIOD)
- **Values**: SUM(OBS_VALUE)
- **Scope**: Construct a Country × Year employment matrix serving as the lookup source (reference matrix) for denominator alignment in WRI C10 TOT H and WRI C10 H CIRCUMSTANCE (INDEX–MATCH integration in Appendix G.2).

2. Pivot Act IT – Employment (Italy, Activities A–U)

- **Source**: Employment IT AU
- **Rows**: NACE Rev.2 Activity
- **Columns**: Year
- **Values**: SUM(OBS_VALUE)
- **Scope**: Construct an Activity × Year employment matrix serving as the lookup source for denominator alignment in WRI IT TOT AU (Appendix G.2).

3. Pivot Circ – Circumstance Validation

- **Source**: WRI C10 H CIRCUMSTANCE
- **Rows**: Circumstance
- **Columns**: Year
- **Values**: SUM(OBS_VALUE)
- **Scope**: Validate time-series consistency prior to denominator integration, detect structural breaks, and identify zero-value sequences before gender and employment alignment.

4. Pivot C10 H % GENDER – Female Employment Share (36 Countries, Activity H)

- **Source**: Employment H 36 C
- **Rows**: Country
- **Columns**: Year
- **Values**: Percentage of Female Employees
- **Scope**: Construct a Country × Year gender-employment matrix serving as the lookup source for retrieving female employment shares in WRI C10 TOT H and deriving male shares (1 − female share) in WRI C10 H CIRCUMSTANCE (Appendix G.2, points 3 and 7).

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5. Pivot Employment IT AU Gender % – Male Employment Share (Italy, Activities A–U)

- **Source**: Employment IT AU
- **Rows**: NACE Rev.2 Activity
- **Columns**: Year
- **Values**: Percentage of Male Employees
- **Scope**: Construct an Activity × Year gender-employment matrix serving as the lookup source for denominator alignment in WRI IT TOT AU (Appendix G.2, point 5).

6. Pivot Table % Gender Circ – Female Casualty Share (10 Countries)

- **Source**: WRI C10 TOT H
- **Rows**: Country
- **Columns**: Year (10-year period)
- **Values**: Percentage of Female Casualties
- **Scope**: Construct a Country-level casualty gender-share matrix serving as the lookup source for deriving male casualty shares (1 − female share) in WRI C10 H CIRCUMSTANCE via XLOOKUP (Appendix G.2, point 4).

7. Pivot Gender % Cas IT AU – Female Casualty Share (Italy, Activities A–U)

- **Source**: WRI IT TOT AU
- **Rows**: NACE Rev.2 Activity
- **Columns**: Year
- **Values**: Percentage of Female Casualties
- **Scope**: Construct an Activity × Year casualty gender-share matrix serving as the lookup source for gender-specific casualty allocation in WRI IT TOT AU (Appendix G.2, point 6).

8. Pivot C10 CAS % F – Female Casualty Share (10 Countries, Activity H)

- **Source**: WRI C10 TOT H
- **Rows**: Country
- **Columns**: Year (10-year period)
- **Values**: Percentage of Female Casualties
- **Scope**: Construct a country-level female casualty share matrix serving as the lookup source for cross-country gender incidence comparison in WRI C10 TOT H via XLOOKUP (Appendix G.2, point 8).

G.2 Lookup & Integration Formulas (Denominator Alignment)

1. Employment Merge (Cross-Country Totals)

- **Worksheet**: WRI C10 TOT H
- **New column**: Employment H
- **Formula**: =INDEX('Pivot Emp'!$B$3:$K$27, MATCH([@Country], 'Pivot Emp'!$A$3:$A$27, 0), MATCH([@Year], 'Pivot Emp'!$B$2:$K$2, 0))
- **Scope**: Two-dimensional lookup (Country × Year) to attach the employment denominator to total casualties for cross-country normalized indicators.

2. Employment Merge (Italy A–U)

- **Worksheet**: WRI IT TOT AU
- **New column**: Employment AU
- **Formula**: =INDEX('Pivot Act IT'!$B$5:$K$25, MATCH([@Activity], 'Pivot Act IT'!$A$5:$A$25, 0), MATCH([@Year], 'Pivot Act IT'!$B$4:$K$4, 0))
- **Scope**: Two-dimensional lookup (Activity × Year) to align Italian employment denominators with activity-level casualty totals.

3. Gender Denominator Alignment (Circumstance-Level, Cross-Country)

- **Worksheet**: WRI C10 H CIRCUMSTANCE
- **New column**: % Males/Year
- **Formula**: =1-INDEX('Gender % H C10'!$C$2:$L$11, MATCH([@Country], 'Gender % H C10'!$A$2:$A$11, 0), MATCH([@Year], 'Gender % H C10'!$C$1:$L$1, 0))
- **Scope**: Derive male employment shares (as complement to female share) by Country × Year for gender-specific normalization at record level.

4. Gender Composition (Circumstance-Level, Casualties Share)

- **Worksheet**: WRI C10 H CIRCUMSTANCE
- **New column**: % Male/Casualties
- **Formula**: =1-XLOOKUP([@Country], 'Table % Gender Circ'!$A$2:$A$11, 'Table % Gender Circ'!$C$2:$C$11, "NOT FOUND", 0)
- **Scope**: Compute the male share of casualties (complement to female share) for country-level gender allocation and validation.

5. Gender Denominator Alignment (Italy A–U, Employment Share)

- **Worksheet**: WRI IT TOT AU
- **New column**: % Males/Year
- **Formula**: =INDEX('Employment IT AU Gender %'!$C$2:$L$22, MATCH([@Activity], 'Employment IT AU Gender %'!$A$2:$A$22, 0), MATCH([@Year], 'Employment IT AU Gender %'!$C$1:$L$1, 0))
- **Scope**: Retrieve male employment shares by Activity × Year to support gender-specific denominator construction within Italy.

6. Gender Composition (Italy A–U, Casualties Share)

- **Worksheet**: WRI IT TOT AU

- **New column**: % F Cas./Y
- **Formula**: =INDEX('Gender % Cas IT AU'!$C$2:$L$22, MATCH([@Activity], 'Gender % Cas IT AU'!$A$2:$A$22, 0), MATCH([@Year], 'Gender % Cas IT AU'!$C$1:$L$1, 0))
- **Scope**: Attach female casualty shares by Activity × Year for gender-specific casualty allocation and consistency checks.

7. Gender Denominator Alignment (Cross-Country Totals, Female Employment Share)

- **Worksheet**: WRI C10 TOT H
- **New column**: % F/Year
- **Formula**: =INDEX('C10 H % GENDER'!$C$2:$L$11, MATCH([@Country], 'C10 H % GENDER'!$A$2:$A$11, 0), MATCH([@Year], 'C10 H % GENDER'!$C$1:$L$1, 0))
- **Scope**: Retrieve female employment shares by Country × Year to compute gender-specific employment denominators and normalized rates.

8. Gender Composition (Cross-Country Totals, Female Casualties Share)

- **Worksheet**: WRI C10 TOT H
- **New column**: % Cas F
- **Formula**: =XLOOKUP([@Country], 'C10 CAS % F'!$A$2:$A$11, 'C10 CAS % F'!$C$2:$C$11, "NOT FOUND", 0)
- **Scope**: Attach female casualty shares for cross-country gender incidence comparisons and validation.

G.3 Derived Indicators

1. Casualties per 1,000 Employees

- **New column type description:** Casualties per 1,000 employees.
- **Formula**: =OBS_VALUE / Employees
- **Scope**: Compute normalized incidence rate.
- **Note**: Employment expressed in thousands → direct division produces rate per 1,000 workers.

2. Casualties per 1,000 Male Employees

- **New column type description:** Male casualties per 1,000 male employees.
- **Formula**: = Casualties assigned to the male gender / Employees assigned to the male gender
- **Scope**: Compute normalized incidence rate of male gender.
- **Note**: Employment expressed in thousands → direct division produces rate per 1,000 workers.

3. Casualties per 1,000 Female Employees

- **New column type description:** Male casualties per 1,000 female employees.
- **Formula**: = Casualties assigned to the female gender / Employees assigned to the female gender
- **Scope**: Compute normalized incidence rate of female gender.
- **Note**: Employment expressed in thousands → direct division produces rate per 1,000 workers.

4. Gender Redistribution (Unknown Category)

Employment Share Calculation

Male Share = Male Employment / Total Employment
Female Share = Female Employment / Total Employment

Unknown Redistribution

Male Casualties = Unknown × Male Share
Female Casualties = Unknown × Female Share

Scope

Reallocate “Unknown” gender records proportionally to preserve aggregate totals while enabling gender-disaggregated indicators.

G.4 Backcasting (Structural Missing Data)

Worksheet: EST CRITERIA

Average Share Calculation

Country Share = Country Casualties / Total Casualties

Historical Estimate

Estimated Value = Total (Year t) × Average Share

Scope

- Reconstruct early missing values for aggregate continuity only.
- All estimated rows flagged: Estimation = TRUE.

G.5 Integrity Controls

- All derived fields are formula-driven (no manual overwriting).
- Original observed values preserved.
- Estimated observations appended, not substituted.
- Final curated CSV exports retain explicit calculated fields.

Appendix H – Data Processing Workflow

The analytical dataset was constructed through a staged workflow comprising EUROSTAT data extraction, Excel-based preparation and harmonization, generation of curated CSV datasets, relational structuring in SQLite, and the production of analytical indicators and comparative outputs.

EUROSTAT Extraction → Excel-Based Harmonization → Curated CSV Layer → SQLite Modeling → Analytical Outputs